Complex Numbers

Chapter Overview

1. Understand and manipulate complex numbers

Write $(4+i)/(3-i)$ in the form $a+bi$.

2. Solve quadratics over the complex numbers

Solve $x^2+3x+5=0$.

3. Solve cubic and quartic equations

Given $-2+i$ is a root of $x^3+3x^2+x-5=0$, determine the other roots.

Why complex numbers? 3

The quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

has no real value when the discriminant is negative. Introducing i, with $i^2 = -1$, lets us express these roots.

Complex numbers arose in sixteenth-century work on polynomial equations, including Cardano's work.



$i = \sqrt{-1}$.

An imaginary number has form bi, $b \in \mathbb{R}$, such as $i, 3i, -2i, \pi i$.

A complex number has form $a + bi$, $a, b \in \mathbb{R}$, such as $1 + i, 3 - 2i$. Its real part is a and its imaginary part is b.

Notation

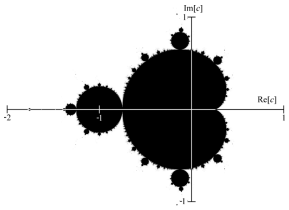
Coefficients usually precede constants and variables, as in $2\pi x$ or $3\pi i x$. Both $a + bi$ and $a + ib$ are valid: multiplication commutes.

The conventional form of Euler's formula is $e^{i\theta} = \cos\theta + i\sin\theta$, with special case $e^{i\pi} + 1 = 0$. Writing $e^{\theta i}$ is mathematically equivalent.

Other Practical Applications 4

1. Fractals

The Mandelbrot set is a famous fractal. We will investigate its recurrence later in this chapter.



2. Analytic number theory

Number theory studies integers. Analytic methods use real and complex variables; for example, the Riemann zeta function accepts complex inputs and is linked to the distribution of primes.

3. Physics and engineering

Applications include signal analysis, quantum mechanics, fluid dynamics, relativity and control theory.

Complex Number Basics 5

Write in terms of i:

$\sqrt{-36}$

$\sqrt{-4}$

$\sqrt{-7}$

Simplify:

$(2+3i)+(4+i)$

4

$i-3(2-i)$

$\frac{10+4i}{2}$

Solving Quadratic Equations 6

Solve $z^2+25=0$.

Solve $z^2+3z+5=0$.

Notation note

We often use x for real variables, n for integers and z for complex variables.

Exercises 1A-1B Pearson Core Pure 1, p. 3, 4-5

4. Write in the form $a + bi$, with a, b simplified surds.

a. $(4 - 2i)/\sqrt{2}$

b. $(2 - 6i)/(1 + \sqrt{3})$

4a

4b

6. Given $z_1 = a + 9i, z_2 = -3 + bi$ and $z_2 - z_1 = 7 + 2i$, find real a, b.

6

Exercises 1A Pearson Core Pure 1, p. 3

7. $z_1 = 4 + i, z_2 = 7 - 3i$. Find, in the form $a + bi$: (a) $z_1 - z_2$, (b) $4z_2$, (c) $2z_1 + 5z_2$.

7a

7b

7c

8. Given $z = a + bi, w = a - bi, a, b \in \mathbb{R}$, show (a) $z + w$ is real, (b) $z - w$ is imaginary.

8a

8b

Exercises 1B Pearson Core Pure 1, p. 4-5 9

2. Solve, giving answers in the form $a + bi$.

a. $(z-3)^2 - 9 = -16$

2a

b. $2(z-7)^2 + 30 = 6$

2b

c. $16(z+1)^2 + 11 = 2$

2c

6. $z^2 + bz + 11 = 0$, $b \in \mathbb{R}$, has distinct non-real roots. Find the possible values of b.

6

Multiplying Complex Numbers 10

$i^2 = -1$. Expand into $a + bi$:

1. $(2+3i)(3-2i)$

1

2. $(5-3i)^2$

2

Find $i^3, i^4, i^{101}, (3i)^5$.

The powers repeat every four: $i, -1, -i, 1$. For i^k, use the remainder when k is divided by 4.

6. Power 101

7. Scaled power

Test Your Understanding 11

Edexcel FP1 June. Given $z = 2 - 3i$, show $z^2 = -5 - 12i$.

Edexcel FP1 June 2010. Expand and simplify $(1 + i)^3$.

Exercise 1C Pearson Core Pure 1, p. 6 12

3. Given $(a + 3i)(1 + bi) = 25 - 39i$, find two possible real pairs (a, b).

1. Equate

2. Solve

3. Pairs

Exercise 1C Pearson Core Pure 1, p. 6 — 13

5. Express $(1+i)^6$ in the form $a-bi$, where a, b are integers.

Use the binomial expansion:

$(a+b)^6 = a^6 + 6a^5b + 15a^4b^2 + 20a^3b^3$
$+15a^2b^4 + 6ab^5 + b^6$.

5

Exercise 1C Pearson Core Pure 1, p. 6 — 14

7. $f(z) = 2z^2 - z + 8$. Find (a) $f(2i)$, (b) $f(3-6i)$.

7a

7b

8. For $f(z) = z^2 - 2z + 17$, show $z = 1 - 4i$ solves $f(z) = 0$.

8

For Interest: The Mandelbrot Set 15

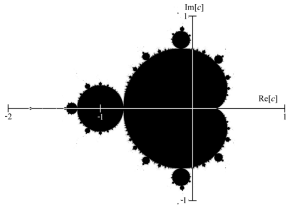
How does the Mandelbrot set work?

Recurrences: $x_{n+1} = x_n + 2, x_1 = 0$ gives $0, 2, 4, \ldots$, diverging.

$x_{n+1} = 5 + x_n/2, x_1 = 0$ gives $0, 5, 7.5, 8.75, \ldots$, converging to 10.

For the Mandelbrot set, use

$$z_{n+1} = z_n^2 + c, \quad z_1 = 0.$$



For $c = 1$:

$z_2 = 0^2 + 1 = 1,$

$z_3 = 1^2 + 1 = 2,$

$z_4 = 2^2 + 1 = 5.$

The sequence is unbounded.

For $c = i$:

$z_2 = 0^2 + i = i,$

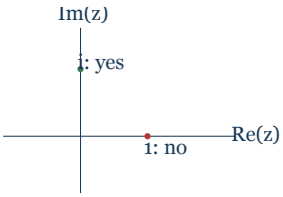
$z_3 = i^2 + i = -1 + i,$

$z_4 = (-1 + i)^2 + i$

$= 1 - 2i - 1 + i = -i,$

$z_5 = (-i)^2 + i = -1 + i.$

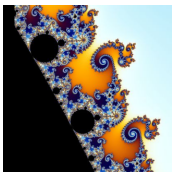
It then alternates and is bounded.



Plot $a + bi$ at (a, b) on an Argand diagram (chapter 2). The Mandelbrot set consists of the values of c whose sequence stays bounded. Bounded sequences need not converge or repeat.

Thus i, plotted at $(0, 1)$, belongs to the set; 1, at $(1, 0)$, does not.

Considering every c produces the fractal boundary. Colours outside the set can show escape speed; black indicates bounded behaviour in this illustration.



Complex Conjugation 16

Let $x = 3 + \sqrt{2}, x^* = 3 - \sqrt{2}$. Here the star labels the algebraic surd conjugate, not the complex conjugate: both numbers are real. Find their sum and product.

Both are rational and free of surds. Their product is useful when rationalising a denominator.

Let $z = 3 + 2i, z^* = 3 - 2i$. Find their sum and product.

4. Complex sum

5. Complex product

Both results are real. This gives a way to make a denominator real, allowing division of complex numbers.

Complex Conjugation 17

If $z = a + bi$, its complex conjugate is $z^* = a - bi$.

Write $(5 + 4i)/(2 - 3i)$ in the form $a + bi$.

2. Division

Multiply numerator and denominator by the conjugate of the denominator, just as for surds.

Remember:

$(a + bi)(a - bi) = a^2 + b^2$.

Calculator tips from the source

On the calculator shown in the original course, choose Mode → Complex, use ENG for i, and OPTN for conjugate, real-part and imaginary-part functions. Menu names vary by model. Complex mode can check products, quotients and powers.

Test Your Understanding 18

FP1 January 2009 Q9(a).

Given $z_1 = 3 + 2i$, $z_2 = (12 - 5i)/z_1$, find z_2 in the form $a + ib$, $a, b \in \mathbb{R}$.

9(a)

Exercise 1D Pearson Core Pure 1, p. 7-8 19

2. Find $z + z^*$ and zz^* for each value.

a. $z = 6 - 3i$

2a

b. $z = 10 + 5i$

2b

c. $z = 3/4 + i/4$

2c

d. $z = \sqrt{5} - 3\sqrt{5}i$

2d

4. Write $(3 - 4i)^2/(1 + i)$ in the form $x + iy$.

4

Exercise 1D Pearson Core Pure 1, p. 7-8 20

6. Given $(5 + 2i)/z = 2 - i$, find z in the form $a + bi$.

6

Edexcel FP1 June 2013 Q9.

For $z = x + iy$, $x, y \in \mathbb{R}$, solve $(3 - i)z^* + 2iz = 9 - i$.

Exam

Roots of Quadratics 21

$x^2 + 4x - 5 = 0$ factors as $(x+5)(x-1) = 0$, so its roots are $-5, 1$. Conversely, those roots give the two factors by the factor theorem.

If α, β are the roots, the monic quadratic is $(x-\alpha)(x-\beta) = x^2 - (\alpha+\beta)x + \alpha\beta$.

For a polynomial with real coefficients, non-real roots occur in complex conjugate pairs.

If a real-coefficient quadratic has root $3 + 4i$, what is its other root?

The non-real condition matters: two real roots need not be equal.

Why Non-real Roots Come in Conjugate Pairs 22

Prove non-real roots of a quadratic with real coefficients occur in conjugate pairs.

Proof 1: quadratic formula

For real a, b, c, $a \neq 0$, suppose $b^2 - 4ac < 0$.

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$= -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$
$$= -\frac{b}{2a} \pm \frac{\sqrt{4ac - b^2}}{2a} i.$$

The real parts are equal and imaginary parts opposite: the roots are conjugates.

Proof 2: sum and product

$\alpha + \beta$ and $\alpha\beta$ must be real. Put $\alpha = a_1 + b_1 i, \beta = a_2 + b_2 i$.

$\alpha + \beta = (a_1 + a_2) + (b_1 + b_2)i$. From the sum, $b_1 + b_2 = 0$, so $b_2 = -b_1$.

$\alpha\beta = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i$.

Its imaginary part vanishes:

$-a_1 b_1 + a_2 b_1 = b_1(a_2 - a_1) = 0$.

Either $b_1 = b_2 = 0$ (both roots real), or $a_2 = a_1$ and $b_2 = -b_1$, proving conjugacy for non-real roots.

Example Question 23

Given $\alpha = 7 + 2i$ is a root of a quadratic with real coefficients, (a) state the other root β, (b) find the quadratic equation.

a

The sum-and-product approach is shorter; remember $(a + bi)(a - bi) = a^2 + b^2$.

b expansion

b quick method

Test Your Understanding 24

FP1 January 2011 Q4.

$2 - 4i$ is a root of $z^2 + pz + q = 0$, with real p, q.

a. State the other root.

b. Find p, q.

a

b

3. Given $2 + 3i$ is a root of a real-coefficient quadratic, (a) state the other root, (b) find the equation in the form $z^2 + bz + c = 0$.

3a

3b

4. Given $5 - i$ is a root of $z^2 + pz + q = 0$, $p, q \in \mathbb{R}$, (a) state the other root, (b) find p, q.

4a

4b

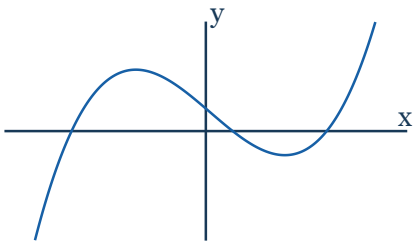
5. Given $-5 + 4i$ is a root of $z^2 + bz + c = 0$, $b, c \in \mathbb{R}$, find b, c.

5

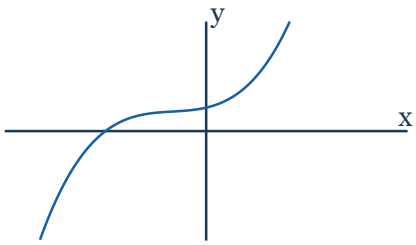
6. Given $1 + 2i$ is a root of a real-coefficient quadratic, find its equation as $z^2 + bz + c = 0$ with integer b, c.

6

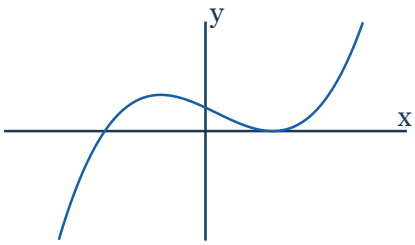
For real-coefficient polynomials, non-real roots occur in conjugate pairs. The fundamental theorem of algebra gives three complex roots counting multiplicity for a cubic.



Comment on the roots.

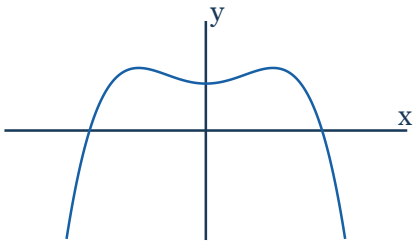


Comment on the roots.

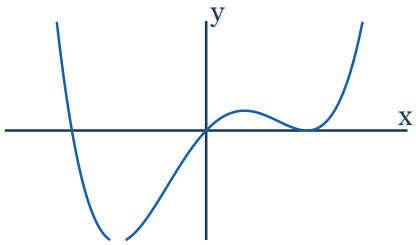


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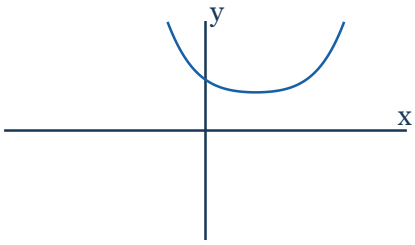
For real-coefficient polynomials, non-real roots occur in conjugate pairs. The fundamental theorem of algebra gives four complex roots counting multiplicity for a quartic.



Comment on the roots.



Comment on the roots.



Comment on the roots.

Example Question 29

Given $3+i$ is a root of

$2z^4-3z^3-39z^2+120z-50=0,$

solve completely.

1. Conjugates

You could also use algebraic long division to obtain the other factor.

2. Factor

3. Solve

Example Question 30

Show z^2+4 is a factor of

$z^4-2z^3+21z^2-8z+68,$

then solve the corresponding equation.

1. Factor

2. Roots

Test Your Understanding 31

FP1 January 2010 Q6.

Given 2 and $5 + 2i$ are roots of
$x^3 - 12x^2 + cx + d = 0$, $c, d \in \mathbb{R}$:

a. State the other non-real root.

b. Find c, d.

a

b

Exercise 1F Pearson Core Pure 1, p. 13-14 32

1. $f(z) = z^3 - 6z^2 + 21z - 26$.

a. Show $f(2) = 0$.

b. Solve $f(z) = 0$ completely.

a

b

Exercise 1F Pearson Core Pure 1, p. 13-14 33

2. $f(z) = 2z^3 + 5z^2 + 9z - 6$.

a. Show $f(1/2) = 0$.

b. Write $f(z) = (2z - 1)(z^2 + bz + c)$, finding real b, c.

c. Solve $f(z) = 0$ using algebra.

b

a

c

Exercise 1F Pearson Core Pure 1, p. 13-14 34

3. $g(z) = 2z^3 - 4z^2 - 5z - 3$. Given $z = 3$ is a root, solve $g(z) = 0$ completely.

2. Solve

1. Division

Exercise 1F Pearson Core Pure 1, p. 13-14 35

4. $p(z) = z^3 + 4z^2 - 15z - 68$. Given $-4 + i$ is a root:

a. Show $z^2 + 8z + 17$ is a factor.

b. Solve $p(z) = 0$.

a

b

Exercise 1F Pearson Core Pure 1, p. 13-14 36

5. $f(z) = z^3 + 9z^2 + 33z + 25$,
$f(z) = (z + 1)(z^2 + az + b)$.

a. Find real a, b.

b. Find all three roots.

c. Find their sum.

a

b

c

6. $g(z) = z^3 - 12z^2 + cz + d$, $c, d \in \mathbb{R}$, has roots 6 and $3 + i$.

a. State the other non-real root.

b. Find c, d.

6a

6b

7. $h(z) = 2z^3 + 3z^2 + 3z + 1$. Given $2z + 1$ is a factor, find all roots.

7

8. $f(z) = z^3 - 6z^2 + 28z + k$, with $f(2) = 0$.

a. Find k.

b. Find the other roots.

a

b

9. Find all four roots of $z^4 - 16 = 0$.

9

10. $f(z) = z^4 - 12z^3 + 31z^2 + 108z - 360$.

a. Write $f(z) = (z^2 - 9)(z^2 + bz + c)$.

b. Hence find all roots.

10a

10b

11. $g(z) = z^4 + 2z^3 - z^2 + 38z + 130$. Given $g(2 + 3i) = 0$, find all roots.

1. Conjugate factor

2. Divide

3. Roots

12. $f(z) = z^4 - 10z^3 + 71z^2 + Qz + 442$, $Q \in \mathbb{R}$, has root $2 - 3i$.

a. Show $z^2 - 6z + 34$ is a factor.

b. Find Q.

c. Solve $f(z) = 0$.

a

b

c